CSC649: SPECIAL TOPICS IN COMPUTER SCIENCE

Assignment 1

Put this page at the front of your PDF report.

Student name	MUHAMMAD FAHDLAN BIN ROSLAN
Student ID	2022899592
Class	NBCS2309A
My problem (one line)	The findings help businesses and marketers understand customer opinions, improve marketing strategies, and make data-driven decisions based on real user feedback.
Specialized technology chosen	Machine Learning-based Sentiment Analysis
GitHub profile link	https://github.com/fahdlan-roslan
Project repository link	https://github.com/fahdlan-roslan/customer-feedback-review
Repository status (Public, or lecturer added)	Public
Did you use GenAI? Which tools?	Copilot
FSKM GenAI declaration submitted? (Yes / No)	Yes

Declaration

I confirm that this work is my own. I have checked all facts and references, I have declared my use of GenAI, and I have followed the FSKM GenAI guide and UiTM academic integrity rules.

Signature: _____ Date: 30/102026

Introduction

Problem Area

Customer feedback is important for businesses and marketers to understand customer satisfaction and improve their services. However, analyzing a large amount of feedback from surveys, reviews, and social media takes a lot of time and can cause important information to be missed. This can affect customer satisfaction, brand reputation, and business decision-making, so Machine Learning can be used to automatically classify feedback and provide useful insights.

Chosen Technology

Machine Learning (Sentiment Analysis)

Repository link

https://github.com/fahdlan-roslan/customer-feedback-review

Part A: Problem Formulation

Problem Statement

Businesses and marketers are affected by the increasing volume of customer feedback collected through surveys, online reviews, and social media platforms. The main issue is that manually reviewing and analyzing large amounts of feedback is time-consuming, inefficient, and prone to human error, causing valuable customer insights and urgent concerns to be overlooked or addressed too late. This matters because unresolved customer issues can reduce customer satisfaction, damage a company's brand reputation, and negatively affect business decisions; for example, online community platforms may receive thousands of customer reviews and comments daily, making it difficult for staff to identify common complaints and emerging issues in a timely manner.

Evidence

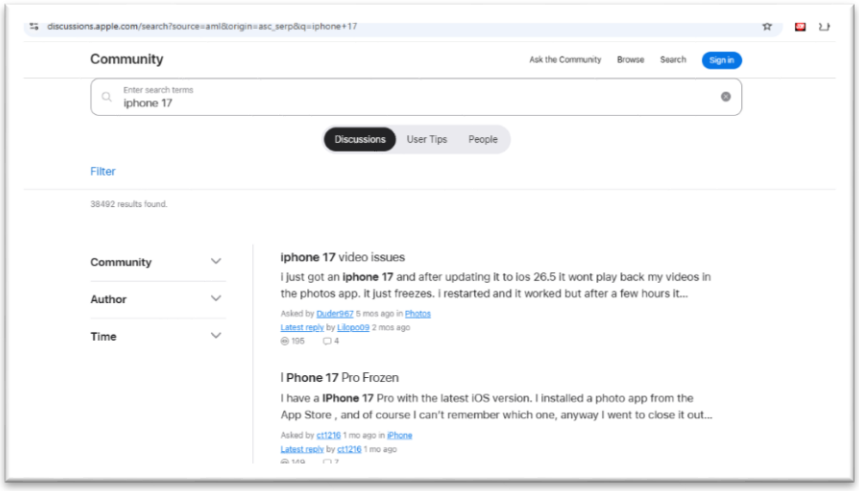


Figure 1. Discussion in Community Apple filter by iPhone 17

Figure 1 show that in official website Apple community, there are a lot of feedback review by customer when filter by iPhone 17 there are around 38492 comments.

Users and Scope

Users	Stakeholders
Business Managers, Marketing Teams, Customer Service Representatives, Business Analysts	Customers, Management, Customer Service Team, IT Department, Business Analysts.

In Scope	Out of Scope
Collection of customer feedback from reviews, surveys, and social media	Responding directly to customer feedback.
Text preprocessing and data cleaning	Customer complaint resolution and case management.
Sentiment analysis using machine learning techniques.	Analysis of voice, image, or video feedback.
Classification of feedback into positive, negative, and neutral categories.	Real-time chatbot functionality.
Generation of sentiment reports and trend summaries.	

Part B: Requirements

Functional requirements

	Requirement
FR1	The system shall allow users to upload customer feedback datasets for analysis.
FR2	The system shall clean and prepare the feedback data by removing irrelevant information and duplicate entries.
FR3	The system shall analyze customer feedback using a machine learning model.
FR4	The system shall determine whether each feedback entry is positive, negative, or neutral.
FR5	The system shall generate a report that summarizes customer sentiment and highlights key findings.

Non-functional requirements

	Requirement
NFR1	The system shall process and classify customer feedback within 2 seconds per submitted record.
NFR2	The system shall achieve at least 80% accuracy in sentiment classification when tested with a labelled dataset.
NFR3	The system shall provide results in a clear and easy to understand format.

User Stories

User Story		Acceptance Criteria
As a business manager, I want to view customer sentiment trends, so that I can make informed decisions to improve products and services.	FR5	1- The system shall generate a report showing the number and percentage of positive, negative, and neutral feedback. 2- The report shall highlight key findings and sentiment trends in an easy-to-understand format.
As a marketer, I want customer feedback to be automatically classified as positive, negative, or neutral, so that I can quickly understand customer opinions without manually reviewing every comment.	FR4	1- The system shall classify every feedback entry into one of three categories: positive, negative, or neutral. 2- The classification result shall be displayed together with the corresponding feedback entry.
As a customer service representative, I want to identify negative feedback quickly, so that I can address customer issues and improve customer satisfaction.	FR3	1- The system shall analyze submitted feedback using a machine learning model and identify negative feedback. 2- The system shall present negative feedback results within 2 seconds per submitted record (NFR1)

Part C: Specialized Technology, Fundamental Concepts

What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every task. Instead of following fixed rules, ML algorithms use historical data to identify relationships and improve their performance over time. In this project, machine learning is used to analyze customer feedback and automatically determine whether the sentiment is positive, negative, or neutral.

Key Concepts

- **Data Collection**: Customer feedback is gathered from sources such as surveys, reviews, and social media comments.
- **Data Preprocessing:** The collected text is cleaned by removing unnecessary characters, duplicate entries, and irrelevant words.
- **Model Training:** A machine learning algorithm learns patterns from previously labeled feedback data.
- **Prediction:** The trained model classifies new customer feedback into sentiment categories such as positive, negative, or neutral.
- **Evaluation:** The model's performance is measured using metrics such as accuracy, precision, and recall.

How It Works

Machine learning first receives customer feedback data as input. The data is cleaned and prepared before being used to train a model. The model learns from examples where the sentiment labels are already known. Once trained, the model can predict the sentiment of new feedback automatically and generate useful insights for businesses.

Customer Feedback	Predicted Sentiment
The product quality is excellent and delivery was fast.	Positive
The service was average and acceptable.	Neutral
I am disappointed with the delayed delivery.	Negative

Frameworks, Models, Methods and Tools

- **Decision Tree:** A Decision Tree is a supervised learning algorithm that classifies data by creating a tree-like structure of decisions. It is easy to understand and useful for predicting categories such as customer sentiment.
  Usage: Customer feedback classification.
- **K-Means Clustering:** K-Means is an unsupervised machine learning and it capable of grouping together similar non-labeled data.
  Usage: Identifying groups of customers with similar feedback patterns.
- **Scikit-learn:** Scikit-learn is a popular Python machine learning library that provides tools for data preprocessing, model training, testing, and evaluation.
  Usage: Building and evaluating sentiment analysis models efficiently.

Fit to Your Problem

How Machine Learning Can Address the Problem

Machine learning is capable of automatically analyzing large volumes of customer feedback and classifying it into positive, neutral, or negative sentiment. This reduces the need for manual review and helps businesses quickly identify customer concerns, trends, and opportunities for improvement. This technology is capable of processing thousands of feedback records in a short amount of time, thereby delivering faster and more consistent results compared to manual analysis.

Benefits

- **Time Efficiency** - Reduces the time required to analyze customer feedback.
- **Scalability** - Capable of handling a large volume of reviews and comments.
- **Improved Decision Making** - Providing insights that support business improvement.
- **Consistency** - Eliminating inconsistencies that caused by human interpretation.

Limitations and Risks

- **Data Quality Dependency** - Machine learning models require large amounts of clean and accurately labeled data. Poor-quality data can reduce model performance.
- **Bias and Accuracy Issues** - The model may produce biased or inaccurate results if the training data does not represent all customer groups or feedback types.
- **Privacy Concerns** - Customer feedback may contain personal information, which requires appropriate data protection and compliance with privacy regulations.

Comparison with an Alternative Approach

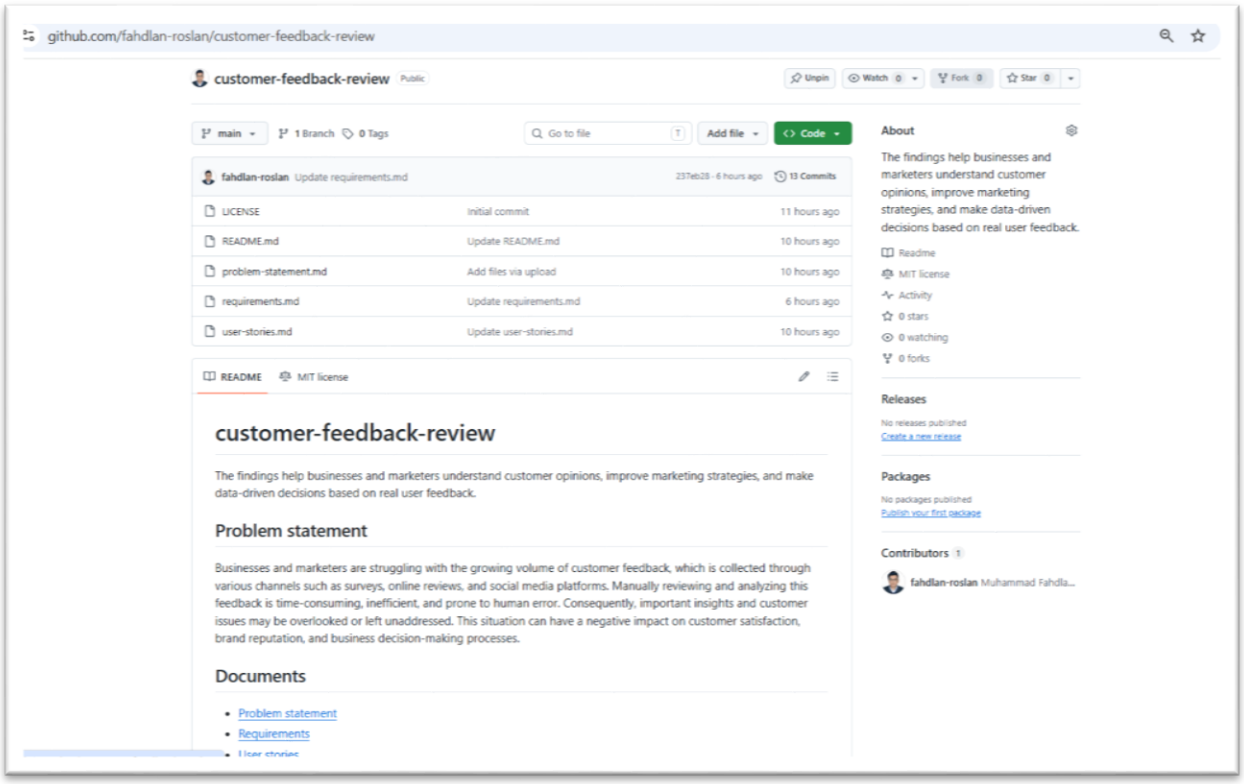
Alternative Approach: Rule-Based Sentiment Analysis

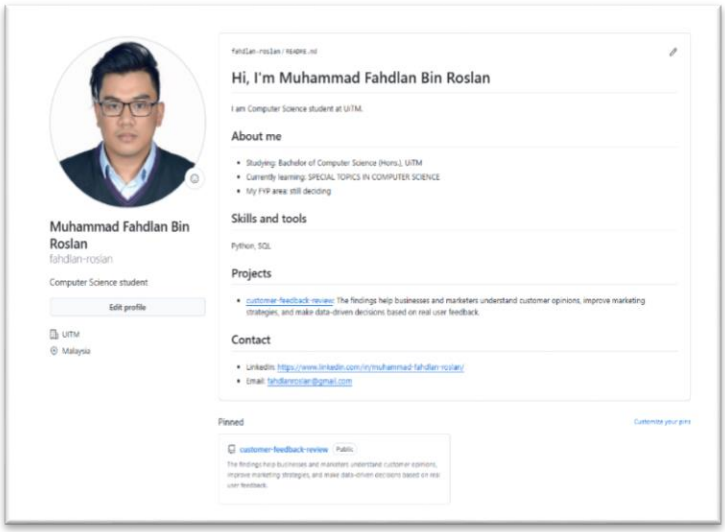
Rule-Based Approach	**Machine Learning Approach**
Uses predefined keywords and rules to determine sentiment	Learn patterns from historical data to make predictions automatically.
Suitable only for simple sentiment analysis with limited data	Highly suitable for analyzing large volumes of customer feedback
Generally higher for large and diverse datasets	Lower when feedback contains complex language or new terms

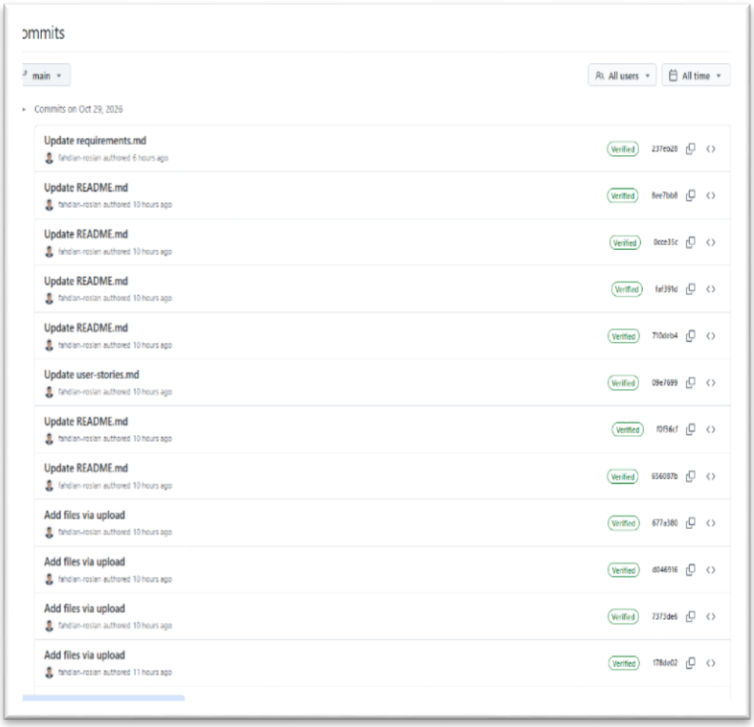
Choice and Justification

Machine Learning was chosen because it is capable of efficiently analyzing large volumes of customer feedback and automatically identifying sentiment patterns. Compared to manual analysis or Rule-Based systems, Machine Learning offers greater accuracy, scalability, and adaptability in handling various types of customer comments. It enables businesses and marketers to gain insights faster, identify customer concerns earlier, and make data-driven decisions. Therefore, Machine Learning is the most suitable technology for addressing the challenge of analyzing customer feedback.

Part D: GitHub Portfolio







REFERENCES

- Tan, Steinbach, Karpatne, Kumar, Lecture Notes, Chapter 1, Introduction to Data Mining, 2nd Edition, 2018
- Pang-Ning Tan, Michael Steinbach & Vipin Kumar, Introduction to Data Mining, Addison Wesley, 2019.
- Jiawei Han and Micheline Kamber, Data Mining: Concepts and Techniques, 3rd Edition, Morgan Kaufmann, 2012.
- Coenen, F. Data mining: past, present and future. Knowledge Engineering Review, 26(1), 25-29, 2011
- Gregory Piatetsky-Shapiro, Data Science: Past, Present, and Future KDnuggets 1© Kdnuggets, 2016

UNIVERSITI TEKNOLOGI MARA | Fakulti Sains Komputer Dan Matematik

GENERATIVE AI TOOLS USAGE DECLARATION

Perisytiharan Penggunaan Alatan Kecerdasan Buatan Generatif (KBG)

STUDENT INFORMATION / MAKLUMAT PELAJAR

Assignment Title / Tajuk Tugasan	Development of a Machine Learning-Based Customer Feedback Analysis System.
Group / Kumpulan	NBCS2309A
Student Name / Nama Pelajar	MUHAMMAD FAHDLAN BIN ROSLAN
Student ID / No. Pelajar	2022899592
Course Instructor / Pensyarah Kursus	DR. TENGKU ZATUL HIDAYAH BINTI TENGKU PETRA
Thesis Supervisor (if applicable) / Penyelia Tesis (jika berkaitan)	-
Thesis Title (if applicable) / Tajuk Tesis (jika berkaitan)	-

STUDENT AI USE DECLARATION / PERISYTIHARAN PENGGUNAAN KBG PELAJAR

I declare that in preparing this assignment/thesis, I / Saya mengisytiharkan bahawa, semasa menyediakan tugasan/tesis ini, saya:

☐ Did not use generative AI tools, except for basic language assistance (e.g. spelling or grammar correction). / Tidak menggunakan alat KBG, kecuali bantuan bahasa asas.

☒ Used generative AI tools (e.g. ChatGPT, Gemini, Copilot, Claude, Scopus AI, ResearchRabbit) for the following purposes / Menggunakan alat KBG untuk tujuan berikut:

☒ **Penjanaan idea / Idea Generation (cth. masalah atau reka bentuk kajian, hipotesis)**
☒ **Carian literatur / Literature Search (cth. meringkaskan, mengenal pasti sumber)**
☐ Reka bentuk kaedah dan eksperimen / Methods and Experiment Design (cth. penyediaan, penalaan model)
☐ Analisis data / Data Analysis (cth. analisis kewangan, coding, tafsiran)
☐ Pembangunan teori / Theoretical Development (cth. analisis konseptual)
☐ Pembangunan kod / Code Development (cth. menjana algoritma, skrip)
☐ Visualisasi dan pembentangan / Visualisation and Presentation (cth. graf, slaid, infografik, rajah, aliran kerja, format)
☐ Penjanaan imej dan video / Image and Video Generation (cth. video, grafik, montaj, imej AI)
☐ Suntingan / Editing (cth. tatabahasa, kebolehbacaan)
☒ **Bantuan penulisan / Writing Assistance (cth. menyusun struktur, parafrasa)**
☐ Format sitasi / Citation Formatting (cth. menyusun rujukan)
☐ Lain-lain / Others (nyatakan dalam jadual alat)

OVERALL GenAI USAGE LEVEL / TAHAP PENGGUNAAN KBG KESELURUHAN

Level 3: Hybrid Collaboration (31%–60%) — Tahap 3: Kolaborasi Hibrid

GenAI contributes part of the content; the student edits, analyses and adapts it.
KBG menyumbang sebahagian kandungan; pelajar menyunting, menganalisis dan menyesuaikan.

GENERATIVE AI TOOLS USED / ALAT KBG YANG DIGUNAKAN

Generative AI Tool / Alat	Purpose of Use / Tujuan	Extent and Justification / Sejauh mana dan justifikasi
Copilot	Used to assist with brainstorming ideas, refining problem statements, drafting user stories, explaining machine learning concepts, and improving the clarity and structure of assignment content.	-

CONFIRMATION / PENGESAHAN

I confirm that this information is true, and I take full responsibility for the accuracy and originality of my work. / Saya mengesahkan maklumat ini benar dan bertanggungjawab sepenuhnya atas ketepatan dan keaslian kerja saya.

Student Name and Signature / Nama dan Tandatangan Pelajar	MUHAMMAD FAHDLAN BIN ROSLAN (tandatangan tulis tangan / handwritten signature)
Supervisor Name and Signature (if applicable) / Nama dan Tandatangan Penyelia	-
Date / Tarikh	30/10/2026